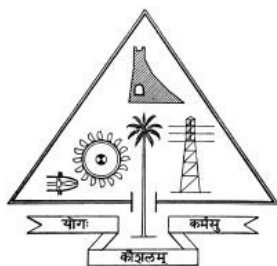


CareChain: A Blockchain-Based Medical Record Exchange System

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submitted by

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ABSTRACT

Imagine this scenario: A patient is rushed to an emergency room unconscious. The doctors have no access to their medical history—no known allergies, no previous surgeries, no ongoing treatments. This critical information gap is not fiction; it's our daily healthcare reality. Today, patient medical records are locked away in separate hospital systems, creating dangerous silos of information. When patients move between hospitals, face emergencies, or seek specialized care, these fragmented records lead to delayed diagnoses, repeated tests, and potentially life-threatening medical decisions. CareChain addresses this fundamental flaw by creating a secure network where medical information can flow safely between authorized healthcare providers while keeping patients firmly in control of their own health data.

Our platform transforms how healthcare providers collaborate. Through an intuitive, consent-based system, patients can securely share their complete medical history—including diagnostic reports, scan images, and treatment records—with doctors across different hospitals. Using blockchain technology, every access request and data transfer creates a permanent, transparent audit trail that patients can monitor in real-time. Meanwhile, advanced encryption and distributed storage ensure sensitive health information remains confidential yet immediately accessible during medical emergencies. CareChain bridges the critical divide between collaborative healthcare and individual privacy, creating an environment where doctors make informed decisions based on complete patient histories while respecting the fundamental right to data sovereignty. This system doesn't just store records; it saves time, reduces medical errors, and most importantly, saves lives by ensuring that critical health information follows patients wherever they receive care.

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